Supplementary Materials: Night-Voyager: Consistent and Efficient Nocturnal Vision-Aided State Estimation in Cross-Modal Maps

This document provides supplementary materials for the "Night-Voyager: Consistent and Efficient Nocturnal Vision-Aided State Estimation in Cross-Modal Maps".

Appendix A
Derivation of state propagation function

In this section, we derive the IMU propagation function for MSC-RIEKF-FDRC. For simplicity, we only investigate the states $\mathbf{X}_{\boldsymbol{I}_i}$ and $\mathbf{B}_i$ since other states are not correlated with the covariance propagation. We first rewrite the dynamic propagation model in (7) as:

$$\mathbf{X}_{\boldsymbol{I}_{i+1}} = \boldsymbol{\Gamma}_i \boldsymbol{\Psi}(\mathbf{X}_{\boldsymbol{I}_i}) \boldsymbol{\Upsilon}_i \tag{52}$$

where $\boldsymbol{\Gamma}_i$, $\boldsymbol{\Psi}(\mathbf{X}_{\boldsymbol{I}_i})$, and $\boldsymbol{\Upsilon}_i$ are formulated as:

$$\boldsymbol{\Gamma}_i = \begin{bmatrix} \mathbf{I}_{3\times3} & \frac{1}{2}\mathbf{g}\delta t^2 & \delta t \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix}, \boldsymbol{\Upsilon}_i = \begin{bmatrix} \Delta\mathbf{R}_i & \Delta\mathbf{p}_i & \Delta\mathbf{v}_i \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix} \tag{53}$$

$$\boldsymbol{\Psi}(\mathbf{X}_{\boldsymbol{I}_i}) = \begin{bmatrix} {}^L\mathbf{R}_{\boldsymbol{I}_i} & {}^L\mathbf{p}_{\boldsymbol{I}_i} + {}^L\mathbf{v}_{\boldsymbol{I}_i}\delta t & {}^L\mathbf{v}_{\boldsymbol{I}_i} \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix}$$

For $\boldsymbol{\Psi}(\mathbf{X}_{\boldsymbol{I}_i})$, it has the following properties:

$$\boldsymbol{\Psi}(\mathbf{X}_1\mathbf{X}_2) = \boldsymbol{\Psi}(\mathbf{X}_1)\boldsymbol{\Psi}(\mathbf{X}_2) \tag{54}$$

$$\boldsymbol{\Psi}(\mathrm{Exp}(\boldsymbol{\xi})) \approx \mathrm{Exp}(\mathbf{F}\boldsymbol{\xi}), \quad \mathbf{F} = \begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \delta t\mathbf{I}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix} \tag{55}$$

Using (3), (79) can be formulated as:

$$\begin{aligned} &\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{I}_{i+1}})\widehat{\mathbf{X}}_{\boldsymbol{I}_{i+1}} = \boldsymbol{\Gamma}_i\boldsymbol{\Psi}(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{I}_i})\widehat{\mathbf{X}}_{\boldsymbol{I}_i})\boldsymbol{\Upsilon}_i \\ &= \boldsymbol{\Gamma}_i\boldsymbol{\Psi}(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{I}_i}))\boldsymbol{\Psi}(\widehat{\mathbf{X}}_{\boldsymbol{I}_i})\boldsymbol{\Upsilon}_i \approx \boldsymbol{\Gamma}_i\mathrm{Exp}(\mathbf{F}\boldsymbol{\xi}_{\boldsymbol{I}_i})\boldsymbol{\Psi}(\widehat{\mathbf{X}}_{\boldsymbol{I}_i})\boldsymbol{\Upsilon}_i \\ &= \mathrm{Exp}(\mathbf{Ad}(\boldsymbol{\Gamma}_i)\mathbf{F}\boldsymbol{\xi}_{\boldsymbol{I}_i})\boldsymbol{\Gamma}_i\boldsymbol{\Psi}(\widehat{\mathbf{X}}_{\boldsymbol{I}_i})\boldsymbol{\Upsilon}_i \\ &= \mathrm{Exp}(\mathbf{Ad}(\boldsymbol{\Gamma}_i)\mathbf{F}\boldsymbol{\xi}_{\boldsymbol{I}_i})\widehat{\mathbf{X}}_{\boldsymbol{I}_{i+1}} \end{aligned} \tag{56}$$

Therefore, we can derive the formulation of $\boldsymbol{\Phi}_{\boldsymbol{II}}$ as:

$$\boldsymbol{\Phi}_{\boldsymbol{II}} = \mathbf{Ad}(\boldsymbol{\Gamma}_i)\mathbf{F} = \begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \frac{1}{2}(\mathbf{g})_\times\delta t^2 & \mathbf{I}_{3\times3} & \mathbf{I}_{3\times3}\delta t \\ (\mathbf{g})_\times\delta t & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix} \tag{57}$$

To derive the relationship between $\boldsymbol{\xi}_{\boldsymbol{I}_{i+1}}$ and $\boldsymbol{\zeta}_{\boldsymbol{I}_i}$, we first rewrite $\boldsymbol{\Upsilon}_i$ as:

$$\boldsymbol{\Upsilon}_i = \widehat{\boldsymbol{\Upsilon}}_i\widetilde{\boldsymbol{\Upsilon}}_i = \begin{bmatrix} \Delta\widehat{\mathbf{R}}_i & \Delta\widehat{\mathbf{p}}_i & \Delta\widehat{\mathbf{v}}_i \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta\widetilde{\mathbf{R}}_i & \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{p}}_i & \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{v}}_i \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix}$$

$$= \widehat{\boldsymbol{\Upsilon}}_i\mathrm{Exp}\left(\begin{bmatrix} \delta\Delta\boldsymbol{\theta}_i \\ \mathbf{J}_l^{-1}(\delta\Delta\boldsymbol{\theta}_i)\Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{p}}_i \\ \mathbf{J}_l^{-1}(\delta\Delta\boldsymbol{\theta}_i)\Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{v}}_i \end{bmatrix}\right) \approx \widehat{\boldsymbol{\Upsilon}}_i\mathrm{Exp}\left(\begin{bmatrix} \delta\Delta\boldsymbol{\theta}_i \\ \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{p}}_i \\ \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{v}}_i \end{bmatrix}\right) \tag{58}$$

where $\Delta\widetilde{\mathbf{R}}_i$, $\Delta\widetilde{\mathbf{p}}_i$ and $\Delta\widetilde{\mathbf{v}}_i$ are respectively formulated as:

$$\begin{aligned} \Delta\widetilde{\mathbf{R}}_i &= \mathrm{Exp}(-\mathbf{J}_r(\widetilde{\boldsymbol{\omega}}_i\delta t)(\delta\mathbf{b}_{\omega_i} + \mathbf{n}_{d\omega})\delta t) = \mathrm{Exp}(\delta\Delta\boldsymbol{\theta}_i) \\ \Delta\widetilde{\mathbf{p}}_i &= \boldsymbol{\Xi}_4(\delta\mathbf{b}_{\omega_i} + \mathbf{n}_{d\omega}) - \boldsymbol{\Xi}_2(\delta\mathbf{b}_{a_i} + \mathbf{n}_{da}) \\ \Delta\widetilde{\mathbf{v}}_i &= \boldsymbol{\Xi}_3(\delta\mathbf{b}_{\omega_i} + \mathbf{n}_{d\omega}) - \boldsymbol{\Xi}_1(\delta\mathbf{b}_{a_i} + \mathbf{n}_{da}) \end{aligned} \tag{59}$$

where $\mathbf{n}_{d\omega} \sim \mathcal{N}(\mathbf{0}_{3\times1}, \frac{\sigma_\omega^2}{\delta t}\mathbf{I}_{3\times3})$ and $\mathbf{n}_{da} \sim \mathcal{N}(\mathbf{0}_{3\times1}, \frac{\sigma_a^2}{\delta t}\mathbf{I}_{3\times3})$ are the discretized noises. The measurements $\widetilde{\boldsymbol{\omega}}_i$ and $\widetilde{\boldsymbol{a}}_i$ are assumed to be constant during propagation, thus we have $\Delta\widehat{\boldsymbol{\theta}}_i = \widetilde{\boldsymbol{\omega}}_i\delta t = \mathrm{Log}(\Delta\widehat{\mathbf{R}}_i)$. Detailed derivation can be referred to [1], [2]. By substituting (58) into (79), we can derive:

$$\begin{aligned} &\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{I}_{i+1}})\widehat{\mathbf{X}}_{i+1} = \boldsymbol{\Gamma}_i\boldsymbol{\Psi}(\mathrm{Exp}(\mathbf{X}_i)\widehat{\boldsymbol{\Upsilon}}_i\widetilde{\boldsymbol{\Upsilon}}_i \\ &\Rightarrow \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{I}_{i+1}}) = \widehat{\mathbf{X}}_{i+1}\widetilde{\boldsymbol{\Upsilon}}_i\widehat{\mathbf{X}}_{i+1}^{-1} = \mathrm{Exp}(\mathbf{Ad}_{\widehat{\mathbf{X}}_{i+1}}\begin{bmatrix} \delta\Delta\boldsymbol{\theta}_i \\ \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{p}}_i \\ \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{v}}_i \end{bmatrix}) \\ &= \mathrm{Exp}(\mathbf{Ad}_{\widehat{\mathbf{X}}_{i+1}}\begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top \end{bmatrix}\begin{bmatrix} \delta\Delta\boldsymbol{\theta}_i \\ \Delta\widetilde{\mathbf{p}}_i \\ \Delta\widetilde{\mathbf{v}}_i \end{bmatrix}) \\ &= \mathrm{Exp}(\mathbf{Ad}_{\widehat{\mathbf{X}}_{i+1}}\begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top \end{bmatrix}\begin{bmatrix} -\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i)\delta t & \mathbf{0}_{3\times3} \\ \boldsymbol{\Xi}_4 & -\boldsymbol{\Xi}_2 \\ \boldsymbol{\Xi}_3 & -\boldsymbol{\Xi}_1 \end{bmatrix}\begin{bmatrix} \delta\mathbf{b}_{\omega_i} + \mathbf{n}_{d\omega} \\ \delta\mathbf{b}_{a_i} + \mathbf{n}_{da} \end{bmatrix}) \end{aligned} \tag{60}$$

Then we can derive $\boldsymbol{\Phi}_{\boldsymbol{IB}}$ and $\mathbf{G}_{\boldsymbol{II}}$ as:

$$\begin{aligned} \boldsymbol{\Phi}_{\boldsymbol{IB}} = \mathbf{G}_{\boldsymbol{II}} &= \mathbf{Ad}_{\widehat{\mathbf{X}}_{i+1}}\begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top \end{bmatrix}\begin{bmatrix} -\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i) & \mathbf{0}_{3\times3} \\ \boldsymbol{\Xi}_4 & -\boldsymbol{\Xi}_2 \\ \boldsymbol{\Xi}_3 & -\boldsymbol{\Xi}_1 \end{bmatrix} \\ &= \begin{bmatrix} -{}^L\widehat{\mathbf{R}}_{I_{i+1}}\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i)\delta t & \mathbf{0}_{3\times3} \\ -({}^L\mathbf{p}_{I_{i+1}})_\times{}^L\widehat{\mathbf{R}}_{I_{i+1}}\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i)\delta t + \widehat{\mathbf{R}}_{I_i}\boldsymbol{\Xi}_4 & -\widehat{\mathbf{R}}_{I_i}\boldsymbol{\Xi}_2 \\ -({}^L\mathbf{v}_{I_{i+1}})_\times{}^L\widehat{\mathbf{R}}_{I_{i+1}}\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i)\delta t + \widehat{\mathbf{R}}_{I_i}\boldsymbol{\Xi}_3 & -\widehat{\mathbf{R}}_{I_i}\boldsymbol{\Xi}_1 \end{bmatrix} \end{aligned} \tag{61}$$

On the other hand, the biases can be discretized as:

$$\mathbf{b}_{\omega_{i+1}} = \mathbf{b}_{\omega_i} + \mathbf{n}_{db\omega}\delta t \tag{62}$$

$$\mathbf{b}_{a_{i+1}} = \mathbf{b}_{a_i} + \mathbf{n}_{dba}\delta t \tag{63}$$

where $\mathbf{n}_{db\omega} \sim \mathcal{N}(\mathbf{0}_{3\times1}, \frac{\sigma_{b\omega}^2}{\delta t}\mathbf{I}_{3\times3})$ and $\mathbf{n}_{dba} \sim \mathcal{N}(\mathbf{0}_{3\times1}, \frac{\sigma_{ba}^2}{\delta t}\mathbf{I}_{3\times3})$ are the discretized noises. We can then derive:

$$\delta\mathbf{b}_{\omega_{i+1}} = \delta\mathbf{b}_{\omega_i} + \mathbf{n}_{db\omega}\delta t \tag{64}$$

$$\delta\mathbf{b}_{a_{i+1}} = \delta\mathbf{b}_{a_i} + \mathbf{n}_{dba}\delta t \tag{65}$$

Therefore, $\mathbf{\Phi}_i^{i+1}$ and $\mathbf{G}_i$ are formulated as:

$$\mathbf{\Phi}_i^{i+1} = \begin{bmatrix} \mathbf{\Phi}_{\boldsymbol{II}} & \mathbf{\Phi}_{\boldsymbol{IB}} \\ \mathbf{0}_{6\times 9} & \mathbf{I}_{6\times 6} \end{bmatrix}, \quad \mathbf{G}_i = \begin{bmatrix} \mathbf{G}_{\mathbf{II}} & \mathbf{0}_{9\times 6} \\ \mathbf{0}_{6\times 6} & \mathbf{I}_{6\times 6}\delta t \end{bmatrix} \tag{66}$$

Appendix B
Derivation of the Jacobian matrices in linearized measurement error functions

A. Odometer Measurement

By replacing the state in error function (13) with (5), the linearized measurement error function is derived as:

$$\begin{aligned}
\mathbf{z}_{o_k} &= \widetilde{\mathbf{y}}_{o_k} - \widehat{\mathbf{y}}_{o_k} \\
&= {}^O\mathbf{R}_I(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^L\widehat{\mathbf{R}}_{I_k})^\top \Big[\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^L\widehat{\mathbf{v}}_{I_k} \\
&\quad + \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})\boldsymbol{\xi}_{\boldsymbol{v}_{LI_k}}\Big] - {}^O\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_k}^\top{}^L\widehat{\mathbf{v}}_{I_k} + \mathbf{n}_{o_k} \\
&\approx {}^O\mathbf{R}_I\widehat{\mathbf{R}}_{I_k}^\top\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^\top\Big[\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^L\widehat{\mathbf{v}}_{I_k} + \boldsymbol{\xi}_{\boldsymbol{v}_{LI_k}}\Big] \\
&\quad - {}^O\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_k}^\top{}^L\widehat{\mathbf{v}}_{I_k} + \mathbf{n}_{o_k} \\
&\approx {}^O\mathbf{R}_I\widehat{\mathbf{R}}_{I_k}^\top\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^\top\boldsymbol{\xi}_{\boldsymbol{v}_{LI_k}} + \mathbf{n}_{o_k} \\
&\approx {}^O\mathbf{R}_I\widehat{\mathbf{R}}_{I_k}^\top\boldsymbol{\xi}_{\boldsymbol{v}_{LI_k}} + \mathbf{n}_{o_k}
\end{aligned} \tag{67}$$

Then the Jacobian matrix of $\mathbf{z}_{o_k}$ with respect to $\boldsymbol{\xi}_k$ is formulated as in (15).

B. Feature Measurement

By replacing the state in error function (17) with (5), the linearized measurement error function is derived as:

$$\begin{aligned}
\mathbf{z}_{f_{j_k}} &= \widetilde{\mathbf{y}}_{f_{j_k}} - \widehat{\mathbf{y}}_{f_{j_k}} \\
&= h\left[{}^C\mathbf{R}_I{}^L\mathbf{R}_{I_{j_k}}^\top({}^L\mathbf{p}_f - {}^L\mathbf{p}_{I_{j_k}}) + {}^C\mathbf{p}_I\right] - \widehat{\mathbf{y}}_{f_{j_k}} + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k}\Big[{}^C\mathbf{R}_I{}^L\mathbf{R}_{I_{j_k}}^\top({}^L\mathbf{p}_f - {}^L\mathbf{p}_{I_{j_k}}) + {}^C\mathbf{p}_I \\
&\quad - {}^C\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_{j_k}}^\top({}^L\widehat{\mathbf{p}}_f - {}^L\widehat{\mathbf{p}}_{I_{j_k}}) - {}^C\mathbf{p}_I\Big] + \mathbf{n}_{f_k} \\
&= \boldsymbol{\mathcal{H}}_{j_k}\Big[{}^C\mathbf{R}_I(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^L\mathbf{R}_{I_{j_k}})^\top\Big[\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})^L\widehat{\mathbf{p}}_f + \\
&\quad \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})\boldsymbol{\xi}_{\boldsymbol{p}_{Lf}} - \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^L\widehat{\mathbf{p}}_{I_{j_k}} - \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})\boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}}\Big] \\
&\quad - {}^C\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_{j_k}}^\top({}^L\widehat{\mathbf{p}}_f - {}^L\widehat{\mathbf{p}}_{I_{j_k}})\Big] + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k}\Big[{}^C\mathbf{R}_I{}^L\mathbf{R}_{I_{j_k}}^\top(\mathbf{I}_{3\times 3} + (\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})_\times)\,[(\mathbf{I}_{3\times 3} \\
&\quad - (\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})_\times)^L\widehat{\mathbf{p}}_f + \boldsymbol{\xi}_{\boldsymbol{p}_{Lf}} - (\mathbf{I}_{3\times 3} + (\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})_\times)^L\widehat{\mathbf{p}}_{I_{j_k}} \\
&\quad - \boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}}\Big] - {}^C\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_{j_k}}^\top({}^L\widehat{\mathbf{p}}_f - {}^L\widehat{\mathbf{p}}_{I_{j_k}})\Big] + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k}{}^C\mathbf{R}_I{}^L\mathbf{R}_{I_{j_k}}^\top\Big[({}^L\widehat{\mathbf{p}}_f)_\times\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}} - ({}^L\widehat{\mathbf{p}}_f)_\times\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}} + \boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}} \\
&\quad - \boldsymbol{\xi}_{\boldsymbol{p}_{Lf}}\Big] + \mathbf{n}_{f_k}
\end{aligned} \tag{68}$$

The Jacobian matrices of $\mathbf{z}_{f_{j_k}}$ with respect to $\boldsymbol{\xi}_k$ and $\boldsymbol{\xi}_{\boldsymbol{p}_{Lf}}$ are formulated as in (18).

C. Streetlight Observation

By replacing the state in error function (17) with (28), the linearized measurement error function is derived as:

$$\begin{aligned}
\mathbf{z}_{s_k} &= \widetilde{\mathbf{y}}_{s_k} - \widehat{\mathbf{y}}_{s_k} \\
&= h\left[{}^C\mathbf{R}_I{}^L\mathbf{R}_{I_k}^\top({}^L\mathbf{R}_{G_k}{}^G\widetilde{\mathbf{c}}_s + {}^L\mathbf{p}_{G_k} - {}^L\mathbf{p}_{I_k}) + {}^C\mathbf{p}_I\right] - \widehat{\mathbf{y}}_{s_k} + \mathbf{n}_{s_k} \\
&\approx \boldsymbol{\mathcal{H}}_{k_k}\Big[{}^C\mathbf{R}_I{}^L\mathbf{R}_{I_k}^\top({}^L\mathbf{R}_{G_k}{}^G\widetilde{\mathbf{c}}_s + {}^L\mathbf{p}_{G_k} - {}^L\mathbf{p}_{I_k}) + {}^C\mathbf{p}_I \\
&\quad - {}^C\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_k}^\top({}^L\widehat{\mathbf{R}}_{G_k}{}^G\widetilde{\mathbf{c}}_s + {}^L\widehat{\mathbf{p}}_{G_k} - {}^L\widehat{\mathbf{p}}_{I_k}) - {}^C\mathbf{p}_I\Big] + \mathbf{n}_{s_k} \\
&= \boldsymbol{\mathcal{H}}_{k_k}\Big[{}^C\mathbf{R}_I(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^L\widehat{\mathbf{R}}_{I_k})^\top\Big[\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})^L\widehat{\mathbf{R}}_{G_k}{}^G\widetilde{\mathbf{c}}_s + \\
&\quad \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})^L\widehat{\mathbf{p}}_{G_k} + \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})\boldsymbol{\xi}_{\boldsymbol{p}_{LG_k}} - \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^L\widehat{\mathbf{p}}_{I_k} \\
&\quad - \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})\boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}}\Big] - {}^C\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_k}^\top({}^L\widehat{\mathbf{R}}_{G_k}{}^G\widetilde{\mathbf{c}}_s + {}^L\widehat{\mathbf{p}}_{G_k} \\
&\quad - {}^L\widehat{\mathbf{p}}_{I_k})\Big] + \mathbf{n}_{s_k} \\
&\approx \boldsymbol{\mathcal{H}}_{k_k}\Big[{}^C\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_k}^\top(\mathbf{I}_{3\times 3} - (\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})_\times)\Big[(\mathbf{I}_{3\times 3} + (\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})_\times)^L\widehat{\mathbf{R}}_{G_k}{}^G\widetilde{\mathbf{c}}_s \\
&\quad + (\mathbf{I}_{3\times 3} + (\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})_\times)^L\widehat{\mathbf{p}}_{G_k} + \boldsymbol{\xi}_{\boldsymbol{p}_{LG_k}} - (\mathbf{I}_{3\times 3} + (\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})_\times)^L\widehat{\mathbf{p}}_{I_k}) \\
&\quad - \boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}}\Big] - {}^C\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_k}^\top({}^L\widehat{\mathbf{R}}_{G_k}{}^G\widetilde{\mathbf{c}}_s + {}^L\widehat{\mathbf{p}}_{G_k} - {}^L\widehat{\mathbf{p}}_{I_k})\Big] + \mathbf{n}_{s_k} \\
&\approx \boldsymbol{\mathcal{H}}_{k_k}^C\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_k}^\top\Big[({}^L\widehat{\mathbf{R}}_{G_k}{}^G\widetilde{\mathbf{c}}_s + {}^L\widehat{\mathbf{p}}_{G_k})_\times(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}} - \boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}}) \\
&\quad + \boldsymbol{\xi}_{\boldsymbol{p}_{LG_k}} - \boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}}\Big] + \mathbf{n}_{s_k}
\end{aligned} \tag{69}$$

Then the Jacobian matrix of $\mathbf{z}_{s_k}$ with respect to $\boldsymbol{\xi}_k$ is formulated as in (29).

D. Prior Pose

By replacing the state in error function (17) with (31), the linearized measurement error function is derived as:

$$\begin{aligned}
\mathbf{z}_{p_k} &= \widetilde{\mathbf{y}}_{p_k} - \widehat{\mathbf{y}}_{p_k} \\
&= \begin{bmatrix} \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top({}^L\mathbf{R}_{G_k}^\top({}^L\mathbf{p}_{I_k} - {}^L\mathbf{p}_{G_k}) - {}^G\mathbf{p}_{I'}) \\ \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top{}^L\mathbf{R}_{G_k}^\top{}^L\mathbf{R}_{I_k}{}^O\mathbf{R}_I^\top\mathbf{e}_3 \end{bmatrix} \\
&\quad - \begin{bmatrix} \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top({}^L\widehat{\mathbf{R}}_{G_k}^\top({}^L\widehat{\mathbf{p}}_{I_k} - {}^L\widehat{\mathbf{p}}_{G_k}) - {}^G\mathbf{p}_{I'}) \\ \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top{}^L\widehat{\mathbf{R}}_{G_k}^\top{}^L\widehat{\mathbf{R}}_{I_k}{}^O\mathbf{R}_I^\top\mathbf{e}_3 \end{bmatrix} + \mathbf{n}_{p_k}
\end{aligned} \tag{70}$$

For the first element of $\mathbf{z}_{p_k}$, we have:

$$\begin{aligned}
z_{p_k,1} &= \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top({}^L\mathbf{R}_{G_k}^\top({}^L\mathbf{p}_{I_k} - {}^L\mathbf{p}_{G_k}) - {}^G\mathbf{p}_{I'}) - \\
&\quad - \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top({}^L\widehat{\mathbf{R}}_{G_k}^\top({}^L\widehat{\mathbf{p}}_{I_k} - {}^L\widehat{\mathbf{p}}_{G_k}) - {}^G\mathbf{p}_{I'}) \\
&= \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top\Big[(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^L\widehat{\mathbf{R}}_{G_k})^\top\Big[(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^L\widehat{\mathbf{p}}_{I_k} \\
&\quad + \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})\boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}} - \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})^L\widehat{\mathbf{p}}_{G_k} - \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})\boldsymbol{\xi}_{\boldsymbol{p}_{LG_k}}\Big] \\
&\quad - {}^G\mathbf{p}_{I'} - {}^L\widehat{\mathbf{R}}_{G_k}^\top({}^L\widehat{\mathbf{p}}_{I_k} - {}^L\widehat{\mathbf{p}}_{G_k}) + {}^G\mathbf{p}_{I'}\Big] + n_{p_k,1} \\
&\approx \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top\Big[{}^L\widehat{\mathbf{R}}_{G_k}^\top(\mathbf{I}_{3\times 3} - (\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})_\times)\,[(\mathbf{I}_{3\times 3} \\
&\quad + (\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})_\times)^L\widehat{\mathbf{p}}_{I_k} + \boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}} - (\mathbf{I}_{3\times 3} + (\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})_\times)^L\widehat{\mathbf{p}}_{G_k} \\
&\quad - \boldsymbol{\xi}_{\boldsymbol{p}_{LG_k}}\Big] - {}^L\widehat{\mathbf{R}}_{G_k}^\top({}^L\widehat{\mathbf{p}}_{I_k} - {}^L\widehat{\mathbf{p}}_{G_k})\Big] + n_{p_k,1} \\
&\approx \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top{}^L\widehat{\mathbf{R}}_{G_k}^\top\Big[({}^L\widehat{\mathbf{p}}_{I_k})_\times\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}} - ({}^L\widehat{\mathbf{p}}_{I_k})_\times\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}} \\
&\quad + \boldsymbol{\xi}_{\boldsymbol{p}_{LG_k}} - \boldsymbol{\xi}_{\boldsymbol{p}_{LG_k}}\Big] + n_{p_k,1}
\end{aligned} \tag{71}$$

For the second element of $\mathbf{z}_{p_k}$, we have:

$$\begin{aligned}
z_{p_k,2} &= \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top{}^L\mathbf{R}_{G_k}^\top{}^L\mathbf{R}_{I_k}{}^O\mathbf{R}_I^\top\mathbf{e}_3 \\
&\quad - \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top{}^L\widehat{\mathbf{R}}_{G_k}^\top{}^L\widehat{\mathbf{R}}_{I_k}{}^O\mathbf{R}_I^\top\mathbf{e}_3 + n_{p_k,2} \\
&= \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})^L\mathbf{R}_{G_k})^\top(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})^L\mathbf{R}_{I_k}) \\
&\quad {}^O\mathbf{R}_I^\top\mathbf{e}_3 - \mathbf{e}_3^{\top O}\mathbf{R}_I{}^G\mathbf{R}_{I'}^\top{}^L\widehat{\mathbf{R}}_{G_k}^\top{}^L\widehat{\mathbf{R}}_{I_k}{}^O\mathbf{R}_I^\top\mathbf{e}_3 + n_{p_k,2}
\end{aligned}$$

$$
\begin{aligned}
&\approx \mathbf{e}_3^\top {}^O\mathbf{R}_I {}^G\mathbf{R}_{I'}^\top {}^L\widehat{\mathbf{R}}_{G_k}^\top (\mathbf{I}_{3\times3} - (\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})_\times)(\mathbf{I}_{3\times3} + (\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})_\times) \\
&\quad {}^L\widehat{\mathbf{R}}_{I_k} {}^O\mathbf{R}_I^\top \mathbf{e}_3 - \mathbf{e}_3^\top {}^O\mathbf{R}_I {}^G\mathbf{R}_{I'}^\top {}^L\widehat{\mathbf{R}}_{G_k}^\top {}^L\widehat{\mathbf{R}}_{I_k} {}^O\mathbf{R}_I^\top \mathbf{e}_3 + n_{p_k,2} \\
&\approx \mathbf{e}_3^\top {}^O\mathbf{R}_I {}^G\mathbf{R}_{I'}^\top {}^L\widehat{\mathbf{R}}_{G_k}^\top ((\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})_\times - (\boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}})_\times) {}^L\widehat{\mathbf{R}}_{I_k} {}^O\mathbf{R}_I^\top \mathbf{e}_3 \\
&\quad + n_{p_k,2} \\
&= \mathbf{e}_3^\top {}^O\mathbf{R}_I {}^G\mathbf{R}_{I'}^\top {}^L\widehat{\mathbf{R}}_{G_k}^\top ({}^L\widehat{\mathbf{R}}_{I_k} {}^O\mathbf{R}_I^\top \mathbf{e}_3)_\times \boldsymbol{\xi}_{\boldsymbol{R}_{LG_k}} - \mathbf{e}_3^\top {}^O\mathbf{R}_I {}^G\mathbf{R}_{I'}^\top \\
&\quad {}^L\widehat{\mathbf{R}}_{G_k}^\top ({}^L\widehat{\mathbf{R}}_{I_k} {}^O\mathbf{R}_I^\top \mathbf{e}_3)_\times \boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}} + n_{p_k,2}
\end{aligned} \tag{72}
$$

Then the Jacobian matrix of $\mathbf{z}_{p_k}$ with respect to $\boldsymbol{\xi}_k$ is formulated as in (32).

Appendix C
Derivation of the variance $\sigma^2_{p,dg}$ and $\sigma^2_{a,dg}$

To derive the formulation of variance in a more compact form, we use the transformation matrix form to represent the pose. Hence (21) can be rewritten as

$$
r_{p,dg} = \|\widetilde{\mathbf{p}}_{s_{d_k}} - h({}^C\mathbf{T}_I {}^L\widehat{\mathbf{T}}_{I_k}^{-1} {}^L\widehat{\mathbf{T}}_{G_k} {}^G\widetilde{\mathbf{c}}_{s_g})\|_2 \tag{73}
$$

The variance is related to ${}^L\widehat{\mathbf{T}}_{I_k}$ and ${}^L\widehat{\mathbf{T}}_{G_k}$. We denote the error state $\boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}$ as $\boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}} = \begin{bmatrix}\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}}^\top & \boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}}^\top\end{bmatrix}^\top$. The Jacobian matrices of $\delta r_{p,dg}$ with respect to $\boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}$ and $\boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}$ are derived as:

$$
\begin{aligned}
\frac{\partial \delta r_{p,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}} &= -\frac{(\widetilde{\mathbf{p}}_{s_{d_k}} - h({}^C\mathbf{T}_I {}^L\widehat{\mathbf{T}}_{I_k}^{-1} {}^L\widehat{\mathbf{T}}_{G_k} {}^G\widetilde{\mathbf{c}}_{s_g}))^\top \boldsymbol{\mathcal{H}}_k \boldsymbol{\Pi} \frac{\partial \delta {}^C\widetilde{\mathbf{c}}_{s_g}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}}}{r_{p,dg}} \\
\frac{\partial \delta {}^C\widehat{\mathbf{c}}_{s_g}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}} &= \lim_{\boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}} \to \mathbf{0}} \frac{({}^C\mathbf{T}_I \mathrm{Exp}(-\boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}) {}^L\widehat{\mathbf{T}}_{I_k}^{-1} - {}^C\mathbf{T}_I {}^L\widehat{\mathbf{T}}_{I_k}^{-1}) {}^L\widehat{\mathbf{T}}_{G_k} {}^G\widetilde{\mathbf{c}}_{s_g}}{\boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}} \\
&= \lim_{\boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}} \to \mathbf{0}} \frac{-{}^C\mathbf{T}_I {}^L\widehat{\mathbf{T}}_{I_k}^{-1} {}^L\widehat{\mathbf{T}}_{G_k} (\mathbf{Ad}({}^L\widehat{\mathbf{T}}_{G_k}^{-1} {}^L\widehat{\mathbf{T}}_{I_k}) \boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}})^\curlywedge {}^G\widetilde{\mathbf{c}}_{s_g}}{\boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}} \\
&= -{}^C\mathbf{T}_I {}^L\widehat{\mathbf{T}}_{I_k}^{-1} {}^L\widehat{\mathbf{T}}_{g_k} {}^G\widetilde{\mathbf{c}}_{s_g}^{\odot} \mathbf{Ad}({}^L\widehat{\mathbf{T}}_{G_k}^{-1} {}^L\widehat{\mathbf{T}}_{I_k}) \\
\frac{\partial \delta r_{p,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}} &= -\frac{(\widetilde{\mathbf{p}}_{s_{d_k}} - h({}^C\mathbf{T}_I {}^L\widehat{\mathbf{T}}_{I_k}^{-1} {}^L\widehat{\mathbf{T}}_{G_k} {}^G\widetilde{\mathbf{c}}_{s_g}))\boldsymbol{\mathcal{H}}_k \boldsymbol{\Pi} \frac{\partial \delta {}^C\widetilde{\mathbf{c}}_{s_g}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}}}{r_{p,dg}} \\
\frac{\partial \delta {}^C\widetilde{\mathbf{c}}_{s_g}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}} &= \lim_{\boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}} \to \mathbf{0}} \frac{{}^C\mathbf{T}_I {}^L\widehat{\mathbf{T}}_{I_k}^{-1} (\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}) {}^L\widehat{\mathbf{T}}_{G_k} {}^G\widetilde{\mathbf{c}}_{s_j} - {}^L\widehat{\mathbf{T}}_{G_k} {}^G\widetilde{\mathbf{c}}_{s_j})}{\boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}} \\
&= \lim_{\boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}} \to \mathbf{0}} \frac{{}^C\mathbf{T}_I {}^L\widehat{\mathbf{T}}_{I_k}^{-1} {}^L\widehat{\mathbf{T}}_{G_k} (\mathbf{Ad}({}^L\widehat{\mathbf{T}}_{G_k}^{-1}) \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}})^\curlywedge {}^G\widetilde{\mathbf{c}}_{s_j}}{\boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}} \\
&= {}^C\mathbf{T}_I {}^L\widehat{\mathbf{T}}_{I_k}^{-1} {}^L\widehat{\mathbf{T}}_{G_k} {}^G\widetilde{\mathbf{c}}_{s_g}^{\odot} \mathbf{Ad}({}^L\widehat{\mathbf{T}}_{G_k}^{-1}) \\
\sigma^2_{p,dg} &= \begin{bmatrix}\frac{\partial \delta r_{p,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}} & \frac{\partial \delta r_{p,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}}\end{bmatrix} \mathbf{P}_{LG_k}^{LI_k} \begin{bmatrix}\frac{\partial \delta r_{p,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}^\top} & \frac{\partial \delta r_{p,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}^\top}\end{bmatrix}^\top
\end{aligned} \tag{74}
$$

where ${}^G\widetilde{\mathbf{c}}_{s_g}^{\odot} = \begin{bmatrix}\mathbf{I}_{3\times3} & -({}^G\widetilde{\mathbf{c}}_{s_g})_\times \\ \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3}\end{bmatrix}$ $\boldsymbol{\Pi} = \begin{bmatrix}\mathbf{I}_{3\times3} & \mathbf{0}_{3\times1}\end{bmatrix}$, and $\mathbf{P}_{LG_k}^{LI_k}$ denotes the covariance matrix block related to $\boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}$ and $\boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}$.

The variance of $r_{a,dg}$. Similar to $\sigma^2_{p,dg}$, $\sigma^2_{a,dg}$ is also related to ${}^L\widehat{\mathbf{T}}_{I_k}$ and ${}^L\widehat{\mathbf{T}}_{G_k}$. The Jacobian matrices of $\delta r_{a,dg}$ with respect to $\boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}$ and $\boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}$ are

$$
\begin{aligned}
\frac{\partial \delta r_{a,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}} &= \frac{(\left\lfloor {}^C\widetilde{\mathbf{p}}_{s_{d_k}} \right\rfloor {}^C\widehat{\mathbf{c}}_{s_g})^\top}{\|\left\lfloor {}^C\widetilde{\mathbf{p}}_{s_{d_k}} \right\rfloor {}^C\widehat{\mathbf{c}}_{s_g}\|_2} \left\lfloor {}^C\widetilde{\mathbf{p}}_{s_{d_k}} \right\rfloor (\frac{\mathbf{I}_{3\times3}}{\|{}^C\widehat{\mathbf{c}}_{s_g}\|_2} \\
&\quad - \frac{{}^C\widehat{\mathbf{c}}_{s_g} {}^C\widehat{\mathbf{c}}_{s_g}^\top}{\|{}^C\widehat{\mathbf{c}}_{s_g}\|_2^3}) \boldsymbol{\Pi} \frac{\partial \delta {}^C\widetilde{\mathbf{c}}_{s_g}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}} \\
\frac{\partial \delta r_{a,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}} &= \frac{(\left\lfloor {}^C\widetilde{\mathbf{p}}_{s_{d_k}} \right\rfloor {}^C\widehat{\mathbf{c}}_{s_g})^\top}{\|\left\lfloor {}^C\widetilde{\mathbf{p}}_{s_{d_k}} \right\rfloor {}^C\widehat{\mathbf{c}}_{s_g}\|_2} \left\lfloor {}^C\widetilde{\mathbf{p}}_{s_{d_k}} \right\rfloor (\frac{\mathbf{I}_{3\times3}}{\|{}^C\widehat{\mathbf{c}}_{s_g}\|_2} \\
&\quad - \frac{{}^C\widehat{\mathbf{c}}_{s_g} {}^C\widehat{\mathbf{c}}_{s_g}^\top}{\|{}^C\widehat{\mathbf{c}}_{s_g}\|_2^3}) \boldsymbol{\Pi} \frac{\partial \delta {}^C\widetilde{\mathbf{c}}_{s_g}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}} \\
\sigma^2_{a,dg} &= \begin{bmatrix}\frac{\partial \delta r_{a,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}} & \frac{\partial \delta r_{a,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}}\end{bmatrix} \mathbf{P}_{LG_k}^{LI_k} \begin{bmatrix}\frac{\partial \delta r_{a,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LI_k}}^\top} & \frac{\partial \delta r_{a,dg}}{\partial \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}^\top}\end{bmatrix}^\top
\end{aligned} \tag{75}
$$

Appendix D
MSC-RIEKF-FC

A. State Representation

For MSC-RIEKF-FC, the features are associated with IMU navigation state to form $SE_{2+K}(3)$ group. The system state is defined as:

$$
\begin{aligned}
&\mathbf{X}_k = (\mathbf{X}_{\boldsymbol{I}_k}, \mathbf{B}_k, \mathbf{X}_{\boldsymbol{C}_k}, \mathbf{X}_{\boldsymbol{G}_k}) \\
&\mathbf{X}_{\boldsymbol{I}_k} = \left[\begin{array}{c|ccc} {}^L\mathbf{R}_{I_k} & {}^L\mathbf{p}_{I_k} & {}^L\mathbf{v}_{I_k} & {}^L\mathbf{p}_{f_k} \\ \hline \mathbf{0}_{3\times3} & & \mathbf{I}_{3\times3} & \end{array}\right] \\
&\mathbf{B}_k = \begin{bmatrix}\mathbf{b}_{\omega_k}^\top & \mathbf{b}_{a_k}^\top\end{bmatrix}^\top \\
&\mathbf{X}_{\boldsymbol{C}_k} = (\mathbf{X}_{CP_{k_k}}, \cdots, \mathbf{X}_{CP_{k-c+1_k}}),\ \mathbf{X}_{CP_{j_k}} = \begin{bmatrix}{}^L\mathbf{R}_{I_{j_k}} & {}^L\mathbf{p}_{I_{j_k}} \\ \mathbf{0}_{1\times3} & 1\end{bmatrix} \\
&\mathbf{X}_{\boldsymbol{G}_k} = \begin{bmatrix}{}^L\mathbf{R}_{G_k} & {}^L\mathbf{p}_{G_k} \\ \mathbf{0}_{1\times3} & 1\end{bmatrix}
\end{aligned} \tag{76}
$$

We keep the error states in the same formula with that in MSC-RIEKF-FDRC, i.e., $\boldsymbol{\xi}_k = \begin{bmatrix}\boldsymbol{\xi}_{\boldsymbol{I}_k}^\top & \boldsymbol{\zeta}_{\boldsymbol{B}_k}^\top & \boldsymbol{\xi}_{\boldsymbol{C}_k}^\top & \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}^\top & \boldsymbol{\xi}_{\boldsymbol{p}_{Lf_k}}^\top\end{bmatrix}^\top$. The feature error state $\boldsymbol{\xi}_{\boldsymbol{p}_{Lf_k}}$ in (5) is formulated as:

$$
{}^L\mathbf{p}_{f_k} = \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}}) {}^L\widehat{\mathbf{p}}_{f_k} + \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}}) \boldsymbol{\xi}_{\boldsymbol{p}_{Lf_k}} \tag{77}
$$

Since $\mathbf{X}_{CP_{k_k}}$ is the clone of current pose, (78) is equivalent to:

$$
{}^L\mathbf{p}_{f_k} = \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}}) {}^L\widehat{\mathbf{p}}_{f_k} + \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{p}_{CP_{k_k}}}) \boldsymbol{\xi}_{\boldsymbol{p}_{Lf_k}} \tag{78}
$$

B. State Propagation

We also only investigate the states $\mathbf{X}_{\boldsymbol{I}_i}$ and $\mathbf{B}_i$ for simplicity. Since the feature point is associated with the propagation, the dynamic propagation model in (6) is rewritten as:

$$
\mathbf{X}_{\boldsymbol{I}_{i+1}} = \boldsymbol{\Gamma}_i \boldsymbol{\Psi}(\mathbf{X}_{\boldsymbol{I}_i}) \boldsymbol{\Upsilon}_i \tag{79}
$$

where $\boldsymbol{\Gamma}_i$, $\boldsymbol{\Psi}(\mathbf{X}_{\boldsymbol{I}_i})$, and $\boldsymbol{\Upsilon}_i$ are formulated as:

$$
\boldsymbol{\Gamma}_i = \left[\begin{array}{c|c} \mathbf{I}_{3\times3} & \frac{1}{2}\mathbf{g}\delta t^2\ \delta t\ \mathbf{0}_{3\times1} \\ \hline \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \end{array}\right], \boldsymbol{\Upsilon}_i = \left[\begin{array}{c|c} \Delta\mathbf{R}_i & \Delta\mathbf{p}_i\ \Delta\mathbf{v}_i\ \mathbf{0}_{3\times1} \\ \hline \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \end{array}\right]
$$

$$\boldsymbol{\Psi}(\mathbf{X}_{\boldsymbol{I}_i}) = \left[\begin{array}{c|cccc} {}^L\mathbf{R}_{\boldsymbol{I}_i} & {}^L\mathbf{p}_{\boldsymbol{I}_i} + {}^L\mathbf{v}_{\boldsymbol{I}_i}\delta t & {}^L\mathbf{v}_{\boldsymbol{I}_i} & {}^G\mathbf{p}_f \\ \hline \mathbf{0}_{3\times3} & \multicolumn{3}{c}{\mathbf{I}_{3\times3}} \end{array}\right] \tag{80}$$

Using (3), (79) can be formulated as:

$$\begin{aligned}
&\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{I}_{i+1}})\widehat{\mathbf{X}}_{\boldsymbol{I}_{i+1}} = \boldsymbol{\Gamma}_i\boldsymbol{\Psi}(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{I}_i})\widehat{\mathbf{X}}_{\boldsymbol{I}_i})\boldsymbol{\Upsilon}_i \\
&= \boldsymbol{\Gamma}_i\boldsymbol{\Psi}(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{I}_i}))\boldsymbol{\Psi}(\widehat{\mathbf{X}}_{\boldsymbol{I}_i})\boldsymbol{\Upsilon}_i \approx \boldsymbol{\Gamma}_i\mathrm{Exp}(\mathbf{F}\boldsymbol{\xi}_{\boldsymbol{I}_i})\boldsymbol{\Psi}(\widehat{\mathbf{X}}_{\boldsymbol{I}_i})\boldsymbol{\Upsilon}_i \\
&= \mathrm{Exp}(\mathbf{Ad}(\boldsymbol{\Gamma}_i)\mathbf{F}\boldsymbol{\xi}_{\boldsymbol{I}_i})\boldsymbol{\Gamma}_i\boldsymbol{\Psi}(\widehat{\mathbf{X}}_{\boldsymbol{I}_i})\boldsymbol{\Upsilon}_i \\
&= \mathrm{Exp}(\mathbf{Ad}(\boldsymbol{\Gamma}_i)\mathbf{F}\boldsymbol{\xi}_{\boldsymbol{I}_i})\widehat{\mathbf{X}}_{\boldsymbol{I}_{i+1}}
\end{aligned} \tag{81}$$

where the matrix $\mathbf{F}$ is formulated as:

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \delta t\mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix} \tag{82}$$

By expanding $\mathbf{Ad}(\boldsymbol{\Gamma}_i)\mathbf{F}$, we can obtain:

$$\mathbf{Ad}(\boldsymbol{\Gamma}_i)\mathbf{F} = \begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \frac{1}{2}(\mathbf{g})_\times\delta t^2 & \mathbf{I}_{3\times3} & \mathbf{I}_{3\times3}\delta t & \mathbf{0}_{3\times3} \\ (\mathbf{g})_\times\delta t & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix} = \left[\begin{array}{c|c} \boldsymbol{\Phi}_{\boldsymbol{II}} & \mathbf{0}_{9\times3} \\ \hline \mathbf{0}_{3\times9} & \mathbf{I}_{3\times3} \end{array}\right] \tag{83}$$

To derive the relationship between $\boldsymbol{\xi}_{\boldsymbol{I}_{i+1}}$ and $\boldsymbol{\zeta}_{\boldsymbol{I}_i}$, we first rewrite $\boldsymbol{\Upsilon}_i$ as:

$$\begin{aligned}
\boldsymbol{\Upsilon}_i &= \widehat{\boldsymbol{\Upsilon}}_i\widetilde{\boldsymbol{\Upsilon}}_i \\
&= \left[\begin{array}{c|cccc} \Delta\widehat{\mathbf{R}}_i & \Delta\widehat{\mathbf{p}}_i & \Delta\widehat{\mathbf{v}}_i & \mathbf{0}_{3\times1} \\ \hline \mathbf{0}_{3\times3} & \multicolumn{3}{c}{\mathbf{I}_{3\times3}} \end{array}\right]\left[\begin{array}{c|cccc} \Delta\widetilde{\mathbf{R}}_i & \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{p}}_i & \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{v}}_i & \mathbf{0}_{3\times1} \\ \hline \mathbf{0}_{3\times3} & \multicolumn{3}{c}{\mathbf{I}_{3\times3}} \end{array}\right] \\
&= \widehat{\boldsymbol{\Upsilon}}_i\mathrm{Exp}\left(\begin{bmatrix} \delta\Delta\boldsymbol{\theta}_i \\ \mathbf{J}_l^{-1}(\delta\Delta\boldsymbol{\theta}_i)\Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{p}}_i \\ \mathbf{J}_l^{-1}(\delta\Delta\boldsymbol{\theta}_i)\Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{v}}_i \\ \mathbf{0}_{3\times1} \end{bmatrix}\right) \\
&\approx \widehat{\boldsymbol{\Upsilon}}_i\mathrm{Exp}\left(\begin{bmatrix} \delta\Delta\boldsymbol{\theta}_i \\ \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{p}}_i \\ \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{v}}_i \\ \mathbf{0}_{3\times1} \end{bmatrix}\right)
\end{aligned} \tag{84}$$

Then similar to the procedure in (60), we can obtain:

$$\begin{aligned}
&\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{I}_{i+1}})\widehat{\mathbf{X}}_{i+1} = \boldsymbol{\Gamma}_i\boldsymbol{\Psi}(\mathrm{Exp}(\mathbf{X}_i)\widehat{\boldsymbol{\Upsilon}}_i\widetilde{\boldsymbol{\Upsilon}}_i \\
&\Rightarrow \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{I}_{i+1}}) = \widehat{\mathbf{X}}_{i+1}\widetilde{\boldsymbol{\Upsilon}}_i\widehat{\mathbf{X}}_{i+1}^{-1} \\
&= \mathrm{Exp}(\mathbf{Ad}_{\widehat{\mathbf{X}}_{i+1}}\begin{bmatrix} \delta\Delta\boldsymbol{\theta}_i \\ \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{p}}_i \\ \Delta\widehat{\mathbf{R}}_i^\top\Delta\widetilde{\mathbf{v}}_i \\ \mathbf{0}_{3\times1} \end{bmatrix}) \\
&= \mathrm{Exp}\left(\mathbf{Ad}_{\widehat{\mathbf{X}}_{i+1}}\begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \end{bmatrix}\begin{bmatrix} \delta\Delta\boldsymbol{\theta}_i \\ \Delta\widetilde{\mathbf{p}}_i \\ \Delta\widetilde{\mathbf{v}}_i \end{bmatrix}\right)
\end{aligned} \tag{85}$$

By expanding the expression in the exponential function, the following equation is obtained:

$$\begin{aligned}
&\mathbf{Ad}_{\widehat{\mathbf{X}}_{i+1}}\begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \end{bmatrix}\begin{bmatrix} \delta\Delta\boldsymbol{\theta}_i \\ \Delta\widetilde{\mathbf{p}}_i \\ \Delta\widetilde{\mathbf{v}}_i \end{bmatrix} \\
&= \mathbf{Ad}_{\widehat{\mathbf{X}}_{i+1}}\begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \end{bmatrix}\begin{bmatrix} -\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i) & \mathbf{0}_{3\times3} \\ \boldsymbol{\Xi}_4 & -\boldsymbol{\Xi}_2 \\ \boldsymbol{\Xi}_3 & -\boldsymbol{\Xi}_1 \end{bmatrix} \\
&= \begin{bmatrix} {}^L\widehat{\mathbf{R}}_{I_{i+1}} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ ({}^L\mathbf{p}_{I_{i+1}})_\times{}^L\widehat{\mathbf{R}}_{I_{i+1}} & {}^L\widehat{\mathbf{R}}_{I_{i+1}} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ ({}^L\mathbf{v}_{I_{i+1}})_\times{}^L\widehat{\mathbf{R}}_{I_{i+1}} & \mathbf{0}_{3\times3} & {}^L\widehat{\mathbf{R}}_{I_{i+1}} & \mathbf{0}_{3\times3} \\ ({}^L\mathbf{p}_{f_{i+1}})_\times{}^L\widehat{\mathbf{R}}_{I_{i+1}} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & {}^L\widehat{\mathbf{R}}_{I_{i+1}} \end{bmatrix} \\
&\begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \Delta\widehat{\mathbf{R}}_i^\top \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \end{bmatrix}\begin{bmatrix} -\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i) & \mathbf{0}_{3\times3} \\ \boldsymbol{\Xi}_4 & -\boldsymbol{\Xi}_2 \\ \boldsymbol{\Xi}_3 & -\boldsymbol{\Xi}_1 \end{bmatrix} \\
&= \begin{bmatrix} -{}^L\widehat{\mathbf{R}}_{I_{i+1}}\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i)\delta t & \mathbf{0}_{3\times3} \\ -({}^L\widehat{\mathbf{p}}_{I_{i+1}})_\times{}^L\widehat{\mathbf{R}}_{I_{i+1}}\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i)\delta t + \widehat{\mathbf{R}}_{I_i}\boldsymbol{\Xi}_4 & -\widehat{\mathbf{R}}_{I_i}\boldsymbol{\Xi}_2 \\ -({}^L\widehat{\mathbf{v}}_{I_{i+1}})_\times{}^L\widehat{\mathbf{R}}_{I_{i+1}}\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i)\delta t + \widehat{\mathbf{R}}_{I_i}\boldsymbol{\Xi}_3 & -\widehat{\mathbf{R}}_{I_i}\boldsymbol{\Xi}_1 \\ -({}^L\widehat{\mathbf{p}}_{f_{i+1}})_\times{}^L\widehat{\mathbf{R}}_{I_{i+1}}\mathbf{J}_r(\Delta\widehat{\boldsymbol{\theta}}_i)\delta t & \mathbf{0}_{3\times3} \end{bmatrix} \\
&= \left[\begin{array}{c} \boldsymbol{\Phi}_{\boldsymbol{IB}} \\ \hline \boldsymbol{\Phi}_{\boldsymbol{fB}} \end{array}\right]
\end{aligned} \tag{86}$$

We can find that the propagation of feature point error state is associated with the biases. Organizing the above equations, the error state propagation function of MSC-RIEKF-FC is formulated as:

$$\boldsymbol{\xi}_{i+1} = \begin{bmatrix} \boldsymbol{\xi}_{\boldsymbol{I}_{i+1}} \\ \boldsymbol{\xi}_{\boldsymbol{B}_{i+1}} \\ \boldsymbol{\xi}_{\boldsymbol{p}_{Lf_{i+1}}} \end{bmatrix} = \boldsymbol{\Phi}_i^{i+1}\begin{bmatrix} \boldsymbol{\xi}_{\boldsymbol{I}_i} \\ \boldsymbol{\xi}_{\boldsymbol{B}_i} \\ \boldsymbol{\xi}_{\boldsymbol{p}_{Lf_i}} \end{bmatrix} + \mathbf{G}_i\begin{bmatrix} \mathbf{n}_{dg} \\ \mathbf{n}_{da} \\ \mathbf{n}_{db\omega} \\ \mathbf{n}_{dba} \end{bmatrix} \tag{87}$$

$$= \begin{bmatrix} \boldsymbol{\Phi}_{\boldsymbol{II}} & \boldsymbol{\Phi}_{\boldsymbol{IB}} & \mathbf{0}_{9\times3} \\ \mathbf{0}_{6\times9} & \mathbf{I}_{6\times6} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times9} & \boldsymbol{\Phi}_{\boldsymbol{fB}} & \mathbf{I}_{3\times3} \end{bmatrix}\begin{bmatrix} \boldsymbol{\xi}_{\boldsymbol{I}_i} \\ \boldsymbol{\xi}_{\boldsymbol{B}_i} \\ \boldsymbol{\xi}_{\boldsymbol{p}_{Lf_i}} \end{bmatrix} + \begin{bmatrix} \mathbf{G}_{\boldsymbol{II}} & \mathbf{0}_{9\times6} \\ \mathbf{0}_{6\times9} & \mathbf{I}_{6\times6}\delta t \\ \mathbf{G}_{\boldsymbol{fI}} & \mathbf{0}_{3\times6} \end{bmatrix}\begin{bmatrix} \mathbf{n}_{dg} \\ \mathbf{n}_{da} \\ \mathbf{n}_{db\omega} \\ \mathbf{n}_{dba} \end{bmatrix}$$

To analyze the computation cost, we partition $\boldsymbol{\Phi}_i^{i+1}$ and $\mathbf{G}_i$ into several matrix blocks:

$$\boldsymbol{\Phi}_i^{i+1} \triangleq \begin{bmatrix} \boldsymbol{\Phi}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}} & \mathbf{0}_{15\times3} \\ \boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}} & \mathbf{I}_{3\times3} \end{bmatrix} \quad \mathbf{G}_i \triangleq \begin{bmatrix} \mathbf{G}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}} \\ \mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}} \end{bmatrix} \tag{88}$$

where $\boldsymbol{\Phi}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}} \in \mathbb{R}^{15\times15}$ and $\mathbf{G}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}} \in \mathbb{R}^{15\times15}$ represent the the blocks that are independent of the feature, and $\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}} \in \mathbb{R}^{3\times15}, \mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}} \in \mathbb{R}^{3\times15}$ represent the blocks that are associated with the feature. By using the propagation functions in (10), the state covariance matrix is derived as:

$$\begin{aligned}
\widehat{\mathbf{P}}_{i+1} &= \boldsymbol{\Phi}_i^{i+1}\begin{bmatrix} \widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}} & \widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\boldsymbol{f}} \\ \widehat{\mathbf{P}}_{\boldsymbol{f}\bar{\boldsymbol{f}}} & \widehat{\mathbf{P}}_{\boldsymbol{f}\boldsymbol{f}} \end{bmatrix}\boldsymbol{\Phi}_i^{i+1^\top} + \mathbf{G}_i\mathbf{Q}_d\mathbf{G}_i^\top \\
&= \begin{bmatrix} \boldsymbol{\Phi}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\boldsymbol{\Phi}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}^\top & \boldsymbol{\Phi}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}^\top + \boldsymbol{\Phi}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\boldsymbol{f}} \\ \boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\boldsymbol{\Phi}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}^\top + \widehat{\mathbf{P}}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\boldsymbol{\Phi}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}^\top & \begin{array}{c}\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}^\top + \boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\boldsymbol{f}} \\ +\widehat{\mathbf{P}}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}^\top + \widehat{\mathbf{P}}_{\boldsymbol{f}\boldsymbol{f}}\end{array} \end{bmatrix} \\
&\quad + \begin{bmatrix} \mathbf{G}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\mathbf{Q}_d\mathbf{G}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}^\top & \mathbf{G}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\mathbf{Q}_d\mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}}^\top \\ \mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\mathbf{Q}_d\mathbf{G}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}^\top & \mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\mathbf{Q}_d\mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}}^\top \end{bmatrix}
\end{aligned} \tag{89}$$

where $\widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}} \in \mathbb{R}^{(21+6c)\times(21+6c)}, \widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\boldsymbol{f}} \in \mathbb{R}^{(21+6c)\times3}, \widehat{\mathbf{P}}_{\boldsymbol{f}\bar{\boldsymbol{f}}} \in \mathbb{R}^{3\times(21+6c)}, \widehat{\mathbf{P}}_{\boldsymbol{f}\boldsymbol{f}} \in \mathbb{R}^{3\times3}$ are the block components of covariance matrix $\widehat{\mathbf{P}}_i$. It can be seen that since $\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}$ and $\mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}}$ are not zero matrices like (8), the calculation of $\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}^\top$,

$\widehat{\mathbf{P}}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}^{\top}$, $\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\widehat{\mathbf{P}}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\boldsymbol{\Phi}_{\boldsymbol{f}\bar{\boldsymbol{f}}}^{\top}$, $\mathbf{G}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}\mathbf{Q}_d\mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}}^{\top}$, $\mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\mathbf{Q}_d\mathbf{G}_{\bar{\boldsymbol{f}}\bar{\boldsymbol{f}}}^{\top}$, and $\mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}}\mathbf{Q}_d\mathbf{G}_{\boldsymbol{f}\bar{\boldsymbol{f}}}^{\top}$ is required. If there are l features, computing these terms need a computation of $O(l^2)$ flops.

C. State Update

Since only the feature points are associated with the IMU navigation state instead of relative transformation, the feature measurement-based state update is different from (99) while the derivation of other state update equations keeps the same. According to the frame where the feature points are observed, there are two cases to be considered.

1) Feature points are observed by current clone pose. By replacing the state in error function (17) with (78), the linearized measurement error function is derived as:

$$\begin{aligned}
\mathbf{z}_{f_{k_k}} &= \widetilde{\mathbf{y}}_{f_{k_k}} - \widehat{\mathbf{y}}_{f_{k_k}} \\
&= h\left[{}^{C}\mathbf{R}_I{}^{L}\mathbf{R}_{I_{k_k}}^{\top}({}^{L}\mathbf{p}_f - {}^{L}\mathbf{p}_{I_{k_k}}) + {}^{C}\mathbf{p}_I\right] - \widehat{\mathbf{y}}_{f_{k_k}} + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{k_k}\left[{}^{C}\mathbf{R}_I{}^{L}\mathbf{R}_{I_{k_k}}^{\top}({}^{L}\mathbf{p}_f - {}^{L}\mathbf{p}_{I_{k_k}}) + {}^{C}\mathbf{p}_I\right. \\
&\quad \left. -{}^{C}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_{k_k}}^{\top}({}^{L}\widehat{\mathbf{p}}_f - {}^{L}\widehat{\mathbf{p}}_{I_{k_k}}) - {}^{C}\mathbf{p}_I\right] + \mathbf{n}_{f_k} \\
&= \boldsymbol{\mathcal{H}}_{k_k}\left[{}^{C}\mathbf{R}_I(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})^{L}\mathbf{R}_{I_{k_k}})^{\top}\left[\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})^{L}\widehat{\mathbf{p}}_f + \right.\right. \\
&\quad \left.\mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})\boldsymbol{\xi}_{\boldsymbol{p}_{Lf}} - \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})^{L}\widehat{\mathbf{p}}_{I_{j_k}} - \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})\boldsymbol{\xi}_{\boldsymbol{p}_{CP_{k_k}}}\right] \\
&\quad \left. -{}^{C}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_{j_k}}^{\top}({}^{L}\widehat{\mathbf{p}}_f - {}^{L}\widehat{\mathbf{p}}_{I_{j_k}})\right] + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k}\left[{}^{C}\mathbf{R}_I{}^{L}\mathbf{R}_{I_{j_k}}^{\top}(\mathbf{I}_{3\times3} - (\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})_{\times})\left[(\mathbf{I}_{3\times3}\right.\right. \\
&\quad +(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})_{\times})^{L}\widehat{\mathbf{p}}_f + \boldsymbol{\xi}_{\boldsymbol{p}_{Lf}} - (\mathbf{I}_{3\times3} + (\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})_{\times})^{L}\widehat{\mathbf{p}}_{I_{j_k}} \\
&\quad \left.\left. -\boldsymbol{\xi}_{\boldsymbol{p}_{CP_{k_k}}}\right] - {}^{C}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_{j_k}}^{\top}({}^{L}\widehat{\mathbf{p}}_f - {}^{L}\widehat{\mathbf{p}}_{I_{j_k}})\right] + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k}{}^{C}\mathbf{R}_I{}^{L}\mathbf{R}_{I_{j_k}}^{\top}(\boldsymbol{\xi}_{\boldsymbol{p}_{Lf}} - \boldsymbol{\xi}_{\boldsymbol{p}_{CP_{k_k}}}) + \mathbf{n}_{f_k}
\end{aligned} \tag{90}$$

2) Feature points are observed by historical clone pose. By replacing the state in error function (17) with (78), the linearized measurement error function is derived as:

$$\begin{aligned}
\mathbf{z}_{f_{j_k}} &= \widetilde{\mathbf{y}}_{f_{j_k}} - \widehat{\mathbf{y}}_{f_{j_k}} \\
&= h\left[{}^{C}\mathbf{R}_I{}^{L}\mathbf{R}_{I_{j_k}}^{\top}({}^{L}\mathbf{p}_f - {}^{L}\mathbf{p}_{I_{j_k}}) + {}^{C}\mathbf{p}_I\right] - \widehat{\mathbf{y}}_{f_{j_k}} + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k}\left[{}^{C}\mathbf{R}_I{}^{L}\mathbf{R}_{I_{j_k}}^{\top}({}^{L}\mathbf{p}_f - {}^{L}\mathbf{p}_{I_{j_k}}) + {}^{C}\mathbf{p}_I\right. \\
&\quad \left. -{}^{C}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_{j_k}}^{\top}({}^{L}\widehat{\mathbf{p}}_f - {}^{L}\widehat{\mathbf{p}}_{I_{j_k}}) - {}^{C}\mathbf{p}_I\right] + \mathbf{n}_{f_k} \\
&= \boldsymbol{\mathcal{H}}_{j_k}\left[{}^{C}\mathbf{R}_I(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{j_k}}})^{L}\mathbf{R}_{I_{j_k}})^{\top}\left[\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})^{L}\widehat{\mathbf{p}}_f + \right.\right. \\
&\quad \left.\mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})\boldsymbol{\xi}_{\boldsymbol{p}_{Lf}} - \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{j_k}}})^{L}\widehat{\mathbf{p}}_{I_{j_k}} - \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{j_k}}})\boldsymbol{\xi}_{\boldsymbol{p}_{CP_{j_k}}}\right] \\
&\quad \left. -{}^{C}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_{j_k}}^{\top}({}^{L}\widehat{\mathbf{p}}_f - {}^{L}\widehat{\mathbf{p}}_{I_{j_k}})\right] + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k}\left[{}^{C}\mathbf{R}_I{}^{L}\mathbf{R}_{I_{j_k}}^{\top}(\mathbf{I}_{3\times3} - (\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{j_k}}})_{\times})\left[(\mathbf{I}_{3\times3}\right.\right. \\
&\quad +(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}})_{\times})^{L}\widehat{\mathbf{p}}_f + \boldsymbol{\xi}_{\boldsymbol{p}_{Lf}} - (\mathbf{I}_{3\times3} + (\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{j_k}}})_{\times})^{L}\widehat{\mathbf{p}}_{I_{j_k}} \\
&\quad \left.\left. -\boldsymbol{\xi}_{\boldsymbol{p}_{CP_{j_k}}}\right] - {}^{C}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_{j_k}}^{\top}({}^{L}\widehat{\mathbf{p}}_f - {}^{L}\widehat{\mathbf{p}}_{I_{j_k}})\right] + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k}{}^{C}\mathbf{R}_I{}^{L}\mathbf{R}_{I_{j_k}}^{\top}\left[({}^{L}\widehat{\mathbf{p}}_f)_{\times}\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{j_k}}} - ({}^{L}\widehat{\mathbf{p}}_f)_{\times}\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}}\right. \\
&\quad \left. +\boldsymbol{\xi}_{\boldsymbol{p}_{CP_{j_k}}} - \boldsymbol{\xi}_{\boldsymbol{p}_{Lf}}\right] + \mathbf{n}_{f_k}
\end{aligned} \tag{91}$$

D. Observability Analysis

For simplicity, we do not include the biases in our state vector as they have been proved observable [3] for general motion. In addition, according to [3], both MSC-RIEKF and RIEKF are derived based on the same propagation model and linearized measurement error models with only the measurement Jacobians are calculated using different estimates. Therefore, it is equivalent to analyze the observability of RIEKF, as long as we adjust the linearization points. The final considered state for MSC-RIEKF-FC is given by $\mathbf{X}_k = (\mathbf{X}_{\boldsymbol{I}_k}, \mathbf{X}_{\boldsymbol{G}_k}, {}^{L}\mathbf{p}_{f_k})$. The observability matrix is defined as:

$$\boldsymbol{\mathcal{O}} = \left[\boldsymbol{\mathcal{O}}_1^{\top}\ \boldsymbol{\mathcal{O}}_2^{\top}\ \cdots\ \boldsymbol{\mathcal{O}}_k^{\top}\right]^{\top} = \left[\mathbf{H}_1^{\top}\ \boldsymbol{\Phi}_1^{2^{\top}}\mathbf{H}_2^{\top}\ \cdots\ \boldsymbol{\Phi}_1^{k^{\top}}\mathbf{H}_k^{\top}\right]^{\top} \tag{92}$$

where $\mathbf{H}_k = \left[\mathbf{H}_{o_k}^{\top}\ \mathbf{H}_{f_k}^{\top}\ \mathbf{H}_{s_k}^{\top}\ \mathbf{H}_{p_k}^{\top}\right]^{\top} \in \mathbb{R}^{9\times18}$. To take the k-th element of $\boldsymbol{\mathcal{O}}$ for analysis, $\boldsymbol{\Phi}_1^k$ is formulated as:

$$\begin{aligned}
\boldsymbol{\Phi}_1^k &= \left(\begin{bmatrix}\boldsymbol{\Phi}_{\boldsymbol{II}} & \mathbf{0}_{9\times9} \\ \mathbf{0}_{9\times9} & \mathbf{I}_{9\times9}\end{bmatrix}\right)^k \\
&= \begin{bmatrix}
\mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\
\frac{1}{2}(\mathbf{g})_{\times}\Delta t_k^2 & \mathbf{I}_{3\times3} & \mathbf{I}_{3\times3}\Delta t_k & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\
(\mathbf{g})_{\times}\Delta t_k & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\
\mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\
\mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \\
\mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3}
\end{bmatrix}
\end{aligned} \tag{93}$$

$$\begin{aligned}
\mathbf{H}_k\boldsymbol{\Phi}_1^k &= \begin{bmatrix}
\mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{H}_{13} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\
\mathbf{0}_{2\times3} & \mathbf{H}_{22} & \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} & \mathbf{H}_{26} \\
\mathbf{H}_{31} & \mathbf{H}_{32} & \mathbf{0}_{2\times3} & \mathbf{H}_{34} & \mathbf{H}_{35} & \mathbf{0}_{2\times3} \\
\mathbf{H}_{41} & \mathbf{H}_{42} & \mathbf{0}_{2\times3} & \mathbf{H}_{44} & \mathbf{H}_{45} & \mathbf{0}_{2\times3}
\end{bmatrix} \\
&\quad \begin{bmatrix}
\mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\
\frac{1}{2}(\mathbf{g})_{\times}\Delta t_k^2 & \mathbf{I}_{3\times3} & \mathbf{I}_{3\times3}\Delta t_k & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\
(\mathbf{g})_{\times}\Delta t_k & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\
\mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\
\mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \\
\mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3}
\end{bmatrix} \\
&= \begin{bmatrix}
\mathbf{M}_{11} & \mathbf{0}_{3\times3} & \mathbf{M}_{13} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\
\mathbf{M}_{21} & \mathbf{M}_{22} & \mathbf{M}_{23} & \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} & \mathbf{M}_{26} \\
\mathbf{M}_{31} & \mathbf{M}_{32} & \mathbf{M}_{33} & \mathbf{M}_{34} & \mathbf{M}_{35} & \mathbf{0}_{2\times3} \\
\mathbf{M}_{41} & \mathbf{M}_{42} & \mathbf{M}_{43} & \mathbf{M}_{44} & \mathbf{M}_{45} & \mathbf{0}_{2\times3}
\end{bmatrix}
\end{aligned} \tag{94}$$

where the matrix elements are respectively formulated as:

$$\begin{aligned}
&\mathbf{H}_{13} = {}^{O}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_k}^{\top} \\
&\mathbf{H}_{22} = -\mathbf{H}_{26} = -\boldsymbol{\mathcal{H}}_k{}^{C}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_k}^{\top} \\
&\mathbf{H}_{31} = -\mathbf{H}_{34} = \boldsymbol{\mathcal{H}}_k{}^{C}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_k}^{\top}({}^{L}\widehat{\mathbf{c}}_s)_{\times} \\
&\mathbf{H}_{32} = -\mathbf{H}_{35} = -\boldsymbol{\mathcal{H}}_k{}^{C}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_k}^{\top} \\
&\mathbf{H}_{41} = -\mathbf{H}_{44} = \begin{bmatrix} -\mathbf{e}_3^{\top}{}^{I}\mathbf{R}_O^{\top}{}^{G}\mathbf{R}_{I'}^{\top}{}^{L}\widehat{\mathbf{R}}_{G_k}({}^{L}\widehat{\mathbf{p}}_{I_k})_{\times} \\ -\mathbf{e}_3^{\top}{}^{I}\mathbf{R}_O^{\top}{}^{G}\mathbf{R}_{I'}^{\top}{}^{L}\widehat{\mathbf{R}}_{G_k}^{\top}({}^{L}\widehat{\mathbf{R}}_{I_k}{}^{O}\mathbf{R}_I^{\top}\mathbf{e}_3)_{\times} \end{bmatrix} \\
&\mathbf{H}_{42} = -\mathbf{H}_{45} = \begin{bmatrix} \mathbf{e}_3^{\top}{}^{I}\mathbf{R}_O^{\top}{}^{G}\mathbf{R}_{I'}^{\top}{}^{L}\widehat{\mathbf{R}}_{G_k} \\ \mathbf{0}_{3\times3} \end{bmatrix} \\
&\mathbf{M}_{11} = {}^{O}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_k}^{\top}(\mathbf{g})_{\times}\Delta t_k \qquad \mathbf{M}_{13} = {}^{O}\mathbf{R}_I{}^{L}\widehat{\mathbf{R}}_{I_k}^{\top} \\
&\mathbf{M}_{21} = -\frac{1}{2}\boldsymbol{\mathcal{H}}_k{}^{C}\mathbf{R}_I{}^{L}\mathbf{R}_{I_k}^{\top}(\mathbf{g})_{\times}\Delta t_k^2 \quad \mathbf{M}_{22} = -\boldsymbol{\mathcal{H}}_k{}^{C}\mathbf{R}_I{}^{L}\mathbf{R}_{I_k}^{\top}
\end{aligned}$$

$$\mathbf{M}_{23} = -\boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\mathbf{R}_{I_k}^\top \Delta t_k \quad \mathbf{M}_{26} = \boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\mathbf{R}_{I_k}^\top$$
$$\mathbf{M}_{31} = \boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\widehat{\mathbf{R}}_{I_k}^\top ({}^L\widehat{\mathbf{c}}_s)_\times - \frac{1}{2}\boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\mathbf{R}_{I_k}^\top (\mathbf{g})_\times \Delta t_k^2$$
$$\mathbf{M}_{32} = -\boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\widehat{\mathbf{R}}_{I_k}^\top \quad \mathbf{M}_{33} = -\boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\widehat{\mathbf{R}}_{I_k}^\top \Delta t_k$$
$$\mathbf{M}_{34} = -\boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\widehat{\mathbf{R}}_{I_k}^\top ({}^L\widehat{\mathbf{c}}_s)_\times \quad \mathbf{M}_{35} = \boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\widehat{\mathbf{R}}_{I_k}^\top$$
$$\mathbf{M}_{41} = \mathbf{e}_3^\top {}^I\mathbf{R}_O^\top {}^G\mathbf{R}_{I'}^\top \begin{bmatrix} -{}^L\widehat{\mathbf{R}}_{G_k}({}^L\widehat{\mathbf{p}}_{I_k})_\times + \frac{1}{2}{}^L\widehat{\mathbf{R}}_{G_k}(\mathbf{g})_\times \Delta t_k^2 \\ -{}^L\widehat{\mathbf{R}}_{G_k}^\top ({}^L\widehat{\mathbf{R}}_{I_k} {}^O\mathbf{R}_I^\top \mathbf{e}_3)_\times \end{bmatrix}$$
$$\mathbf{M}_{42} = \begin{bmatrix} \mathbf{e}_3^\top {}^I\mathbf{R}_O^\top {}^G\mathbf{R}_{I'}^\top {}^L\widehat{\mathbf{R}}_{G_k} \\ \mathbf{0}_{3\times3} \end{bmatrix} \mathbf{M}_{43} = \begin{bmatrix} \mathbf{e}_3^\top {}^I\mathbf{R}_O^\top {}^G\mathbf{R}_{I'}^\top {}^L\widehat{\mathbf{R}}_{G_k}\Delta t_k \\ \mathbf{0}_{3\times3} \end{bmatrix}$$
$$\mathbf{M}_{44} = \mathbf{e}_3^\top {}^I\mathbf{R}_O^\top {}^G\mathbf{R}_{I'}^\top \begin{bmatrix} {}^L\widehat{\mathbf{R}}_{G_k}({}^L\widehat{\mathbf{p}}_{I_k})_\times \\ {}^L\widehat{\mathbf{R}}_{G_k}^\top ({}^L\widehat{\mathbf{R}}_{I_k} {}^O\mathbf{R}_I^\top \mathbf{e}_3)_\times \end{bmatrix}$$
$$\mathbf{M}_{45} = \begin{bmatrix} -\mathbf{e}_3^\top {}^I\mathbf{R}_O^\top {}^G\mathbf{R}_{I'}^\top {}^L\widehat{\mathbf{R}}_{G_k} \\ \mathbf{0}_{3\times3} \end{bmatrix} \tag{95}$$

The right nullspace $\mathcal{N}_{FC}$ of the observability matrix can be deduced by spanning the following unobservable subspace:

$$\mathcal{N}_{FC} = span\left(\begin{bmatrix} \mathbf{g}^\top & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} & \mathbf{g}^\top & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} \\ \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix}\right)^\top \tag{96}$$

We can see that MSC-RIEKF-FC maintains the correct dimension of the nullspace without introducing spurious gain.

Appendix E
State representation, state propagation, state update, and observability of MSC-RIEKF-FDN

A. State Representation

For MSC-RIEKF-FDN, the features are not maintained in any Lie group. The system state is defined as:

$$\mathbf{X}_k = (\mathbf{X}_{\boldsymbol{I}_k}, \mathbf{B}_k, \mathbf{X}_{\boldsymbol{C}_k}, \mathbf{X}_{\boldsymbol{G}_k}, {}^L\mathbf{p}_{f_k})$$
$$\mathbf{X}_{\boldsymbol{I}_k} = \left[\begin{array}{c|cc} {}^L\mathbf{R}_{I_k} & {}^L\mathbf{p}_{I_k} & {}^L\mathbf{v}_{I_k} \\ \hline \mathbf{0}_{2\times3} & \multicolumn{2}{c}{\mathbf{I}_{2\times2}} \end{array}\right]$$
$$\mathbf{B}_k = \begin{bmatrix} \mathbf{b}_{\omega_k}^\top & \mathbf{b}_{a_k}^\top \end{bmatrix}^\top$$
$$\mathbf{X}_{\boldsymbol{C}_k} = (\mathbf{X}_{CP_{k_k}}, \cdots, \mathbf{X}_{CP_{k-c+1_k}}),\ \mathbf{X}_{CP_{j_k}} = \begin{bmatrix} {}^L\mathbf{R}_{I_{j_k}} & {}^L\mathbf{p}_{I_{j_k}} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix}$$
$$\mathbf{X}_{\boldsymbol{G}_k} = \begin{bmatrix} {}^L\mathbf{R}_{G_k} & {}^L\mathbf{p}_{G_k} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \tag{97}$$

We keep the error states in the same formula with that in MSC-RIEKF-FDRC, i.e., $\boldsymbol{\xi}_k = \begin{bmatrix} \boldsymbol{\xi}_{\boldsymbol{I}_k}^\top & \boldsymbol{\zeta}_{\boldsymbol{B}_k}^\top & \boldsymbol{\xi}_{\boldsymbol{C}_k}^\top & \boldsymbol{\xi}_{\boldsymbol{T}_{LG_k}}^\top & \boldsymbol{\xi}_{\boldsymbol{p}_{Lf_k}}^\top \end{bmatrix}^\top$. The feature is decoupled from the Lie group and we have:

$${}^L\mathbf{p}_{f_k} = \widehat{\mathbf{p}}_{f_k} + \boldsymbol{\xi}_{\boldsymbol{p}_{Lf_k}} \tag{98}$$

B. State Propagation

The derivation of state propagation is same as that in Section. A.

C. State Update

Since only the feature points are not maintained in Lie group, the feature measurement-based state update is different from (99) while the derivation of other state update equations keeps the same. By replacing the state in error function (17) with (98), the linearized measurement error function is derived as:

$$\begin{aligned}
\mathbf{z}_{f_{j_k}} &= \widetilde{\mathbf{y}}_{f_{j_k}} - \widehat{\mathbf{y}}_{f_{j_k}} \\
&= h\left[{}^C\mathbf{R}_I {}^L\mathbf{R}_{I_{j_k}}^\top ({}^L\mathbf{p}_f - {}^L\mathbf{p}_{I_{j_k}}) + {}^C\mathbf{p}_I\right] - \widehat{\mathbf{y}}_{f_{j_k}} + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k}\left[{}^C\mathbf{R}_I {}^L\mathbf{R}_{I_{j_k}}^\top ({}^L\mathbf{p}_f - {}^L\mathbf{p}_{I_{j_k}}) + {}^C\mathbf{p}_I \right. \\
&\quad \left. -{}^C\mathbf{R}_I {}^L\widehat{\mathbf{R}}_{I_{j_k}}^\top ({}^L\widehat{\mathbf{p}}_f - {}^L\widehat{\mathbf{p}}_{I_{j_k}}) - {}^C\mathbf{p}_I\right] + \mathbf{n}_{f_k} \\
&= \boldsymbol{\mathcal{H}}_{j_k}\left[{}^C\mathbf{R}_I(\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}}) {}^L\mathbf{R}_{I_{j_k}})^\top \left[{}^L\widehat{\mathbf{p}}_f + \boldsymbol{\xi}_{\boldsymbol{p}_{Lf}} \right.\right. \\
&\quad \left. -\mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}}) {}^L\widehat{\mathbf{p}}_{I_{j_k}} - \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})\boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}}\right] - {}^C\mathbf{R}_I {}^L\widehat{\mathbf{R}}_{I_{j_k}}^\top \\
&\quad \left.({}^L\widehat{\mathbf{p}}_f - {}^L\widehat{\mathbf{p}}_{I_{j_k}})\right] + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k}\left[{}^C\mathbf{R}_I {}^L\mathbf{R}_{I_{j_k}}^\top (\mathbf{I}_{3\times3} - (\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})_\times)\left[{}^L\widehat{\mathbf{p}}_f + \boldsymbol{\xi}_{\boldsymbol{p}_{Lf}} \right.\right. \\
&\quad \left. -(\mathbf{I}_{3\times3} + (\boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}})_\times) {}^L\widehat{\mathbf{p}}_{I_{j_k}} - \boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}}\right] - {}^C\mathbf{R}_I {}^L\widehat{\mathbf{R}}_{I_{j_k}}^\top \\
&\quad \left.({}^L\widehat{\mathbf{p}}_f - {}^L\widehat{\mathbf{p}}_{I_{j_k}})\right] + \mathbf{n}_{f_k} \\
&\approx \boldsymbol{\mathcal{H}}_{j_k} {}^C\mathbf{R}_I {}^L\mathbf{R}_{I_{j_k}}^\top \left[({}^L\widehat{\mathbf{p}}_f)_\times \boldsymbol{\xi}_{\boldsymbol{R}_{LI_k}} - \boldsymbol{\xi}_{\boldsymbol{p}_{LI_k}} + \boldsymbol{\xi}_{\boldsymbol{p}_{Lf}}\right] + \mathbf{n}_{f_k}
\end{aligned} \tag{99}$$

D. Observability Analysis

In MSC-RIEKF-FDN, the formulation of Φ_1^k is same as (93). Therefore, the k-th element of the observability matrix is:

$$\begin{aligned}
\mathbf{H}_k\boldsymbol{\Phi}_1^k &= \begin{bmatrix} \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{H}_{13} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{H}_{21} & \mathbf{H}_{22} & \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} & \mathbf{H}_{26} \\ \mathbf{H}_{31} & \mathbf{H}_{32} & \mathbf{0}_{2\times3} & \mathbf{H}_{34} & \mathbf{H}_{35} & \mathbf{0}_{2\times3} \\ \mathbf{H}_{41} & \mathbf{H}_{42} & \mathbf{0}_{2\times3} & \mathbf{H}_{44} & \mathbf{H}_{45} & \mathbf{0}_{2\times3} \end{bmatrix} \\
&\begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \frac{1}{2}(\mathbf{g})_\times \Delta t_k^2 & \mathbf{I}_{3\times3} & \mathbf{I}_{3\times3}\Delta t_k & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ (\mathbf{g})_\times \Delta t_k & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix} \\
&= \begin{bmatrix} \mathbf{M}_{11} & \mathbf{0}_{3\times3} & \mathbf{M}_{13} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{M}_{21} & \mathbf{M}_{22} & \mathbf{M}_{23} & \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} & \mathbf{M}_{26} \\ \mathbf{M}_{31} & \mathbf{M}_{32} & \mathbf{M}_{33} & \mathbf{M}_{34} & \mathbf{M}_{35} & \mathbf{0}_{2\times3} \\ \mathbf{M}_{41} & \mathbf{M}_{42} & \mathbf{M}_{43} & \mathbf{M}_{44} & \mathbf{M}_{45} & \mathbf{0}_{2\times3} \end{bmatrix}
\end{aligned} \tag{100}$$

where $\mathbf{H}_{21} = \boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\mathbf{R}_{I_{j_k}}^\top ({}^L\widehat{\mathbf{p}}_{f_k})_\times$, $\mathbf{M}_{21} = -\frac{1}{2}\boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\mathbf{R}_{I_k}^\top (\mathbf{g})_\times \Delta t_k^2 + \boldsymbol{\mathcal{H}}_k {}^C\mathbf{R}_I {}^L\mathbf{R}_{I_{j_k}}^\top ({}^L\widehat{\mathbf{p}}_{f_k})_\times$ and other elements keep the same formulation as (95). The right nullspace $\mathcal{N}_N$ of the observability matrix can be deduced by spanning the following unobservable subspace:

$$\boldsymbol{\mathcal{N}}_N = \begin{bmatrix} \mathbf{g}^\top & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} & \mathbf{g}^\top ({}^L\widehat{\mathbf{p}}_{f_k})_\times \\ \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix}^\top \tag{101}$$

Appendix F
State representation, state propagation, state update, and observability of MSC-RIEKF-FDR

A. State Representation

The state definition in MSC-RIEKF-FDR is formulated as:

$$\mathbf{X}_k = (\mathbf{X}_{\boldsymbol{L}_k}, \mathbf{X}_{\mathbf{G}_k})$$
$$\mathbf{X}_{\boldsymbol{L}_k} = (\mathbf{X}_{\boldsymbol{I}_k}, \mathbf{B}_k, \mathbf{X}_{\boldsymbol{C}_k})$$
$$\mathbf{X}_{\boldsymbol{I}_k} = \begin{bmatrix} {}^L\mathbf{R}_{I_k} & {}^L\mathbf{p}_{I_k} & {}^L\mathbf{v}_{I_k} \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix}$$
$$\mathbf{X}_{\boldsymbol{G}_k} = \begin{bmatrix} {}^L\mathbf{R}_{G_k} & {}^L\mathbf{p}_{G_k} & {}^L\mathbf{p}_{f_{1_k}} & \cdots & {}^L\mathbf{p}_{f_{K_k}} \\ \mathbf{0}_{1\times3} & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{1\times3} & 0 & 0 & \cdots & 1 \end{bmatrix}$$
$$\mathbf{B}_k = \begin{bmatrix} \mathbf{b}_{\omega_k}^\top & \mathbf{b}_{a_k}^\top \end{bmatrix}^\top$$
$$\mathbf{X}_{\boldsymbol{C}_k} = (\mathbf{X}_{CP_{k_k}}, \cdots, \mathbf{X}_{CP_{k-c+1_k}}),\ \mathbf{X}_{CP_{j_k}} = \begin{bmatrix} {}^L\mathbf{R}_{I_{j_k}} & {}^L\mathbf{p}_{I_{j_k}} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix}$$

The error state keeps the same formulation as (5).

B. State Propagation

The derivation of state propagation is same as that in Section. A.

C. State Update

All the derivation of state update equations keep the same.

D. Observability Analysis

In MSC-RIEKF-FDN, the formulation of Φ_1^k is same as (93). Therefore, the k-th element of the observability matrix is:

$$\mathbf{H}_k\mathbf{\Phi}_1^k = \begin{bmatrix} \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{H}_{13} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{H}_{21} & \mathbf{H}_{22} & \mathbf{0}_{2\times3} & \mathbf{H}_{24} & \mathbf{0}_{2\times3} & \mathbf{H}_{26} \\ \mathbf{H}_{31} & \mathbf{H}_{32} & \mathbf{0}_{2\times3} & \mathbf{H}_{34} & \mathbf{H}_{35} & \mathbf{0}_{2\times3} \\ \mathbf{H}_{41} & \mathbf{H}_{42} & \mathbf{0}_{2\times3} & \mathbf{H}_{44} & \mathbf{H}_{45} & \mathbf{0}_{2\times3} \end{bmatrix}$$
$$\begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \frac{1}{2}(\mathbf{g})_\times\Delta t_k^2 & \mathbf{I}_{3\times3} & \mathbf{I}_{3\times3}\Delta t_k & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ (\mathbf{g})_\times\Delta t_k & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix}$$
$$= \begin{bmatrix} \mathbf{M}_{11} & \mathbf{0}_{3\times3} & \mathbf{M}_{13} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{M}_{21} & \mathbf{M}_{22} & \mathbf{M}_{23} & \mathbf{M}_{24} & \mathbf{0}_{2\times3} & \mathbf{M}_{26} \\ \mathbf{M}_{31} & \mathbf{M}_{32} & \mathbf{M}_{33} & \mathbf{M}_{34} & \mathbf{M}_{35} & \mathbf{0}_{2\times3} \\ \mathbf{M}_{41} & \mathbf{M}_{42} & \mathbf{M}_{43} & \mathbf{M}_{44} & \mathbf{M}_{45} & \mathbf{0}_{2\times3} \end{bmatrix} \tag{102}$$

where $\mathbf{H}_{21} = -\mathbf{H}_{24} = \boldsymbol{\mathcal{H}}_k{}^C\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_k}({}^L\widehat{\mathbf{p}}_{f_k})_\times$, $\mathbf{M}_{21} = -\frac{1}{2}\boldsymbol{\mathcal{H}}_k{}^C\mathbf{R}_I{}^L\mathbf{R}_{I_k}^\top(\mathbf{g})_\times\Delta t_k^2 + \boldsymbol{\mathcal{H}}_k{}^C\mathbf{R}_I{}^L\mathbf{R}_{I_{j_k}}^\top({}^L\widehat{\mathbf{p}}_{f_k})_\times$,$\mathbf{M}_{24} = -\boldsymbol{\mathcal{H}}_k{}^C\mathbf{R}_I{}^L\widehat{\mathbf{R}}_{I_k}({}^L\widehat{\mathbf{p}}_{f_k})_\times$ and other elements keep the same formulation as (95). The right nullspace $\mathcal{N}_{R_1}$ of the observability matrix can be deduced by spanning the following unobservable subspace:

$$\mathcal{N}_{R_1} = span\left(\begin{bmatrix} \mathbf{g}^\top & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} & \mathbf{g}^\top & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} \\ \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix}\right)^\top \tag{103}$$

We can see that When the streetlight-based measurements are available, MSC-RIEKF-FDN maintains the correct dimension of the nullspace without introducing spurious gain. When the streetlight-based measurements and prior pose-based measurements are not available, we have:

$$\mathbf{H}_k\mathbf{\Phi}_1^k = \begin{bmatrix} \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{H}_{13} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{H}_{21} & \mathbf{H}_{22} & \mathbf{0}_{2\times3} & \mathbf{H}_{24} & \mathbf{0}_{2\times3} & \mathbf{H}_{26} \end{bmatrix}$$
$$\begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \frac{1}{2}(\mathbf{g})_\times\Delta t_k^2 & \mathbf{I}_{3\times3} & \mathbf{I}_{3\times3}\Delta t_k & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ (\mathbf{g})_\times\Delta t_k & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix}$$
$$= \begin{bmatrix} \mathbf{M}_{11} & \mathbf{0}_{3\times3} & \mathbf{M}_{13} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{M}_{21} & \mathbf{M}_{22} & \mathbf{M}_{23} & \mathbf{M}_{24} & \mathbf{0}_{2\times3} & \mathbf{M}_{26} \end{bmatrix} \tag{104}$$

All the elements keep the same formulation as (102). The right nullspace $\mathcal{N}_{R_2}$ of the observability matrix can be deduced by spanning the following unobservable subspace:

$$\boldsymbol{\mathcal{N}}_{R_2} = \begin{bmatrix} \mathbf{g}^\top & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} & \mathbf{g}^\top & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} \\ \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & {}^L\widehat{\mathbf{p}}_{L_{f_k}} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \end{bmatrix}^\top \tag{105}$$

The lack of streetlight observations and prior poses results in the insufficient constraints on the relative pose state, thus the dimension of unobservable subspace should be 10. However, the feature estimate ${}^L\widehat{\mathbf{p}}_{L_{f_k}}$ in $\boldsymbol{\mathcal{N}}_{R_2}$ reduces the dimension to 7 and leads to the inconsistency problem.

Appendix G
State representation, state propagation, state update, and observability of MSC-RIEKF-FDRC

A. State Representation

The state definition in MSC-RIEKF-FDRC is formulated as:

$$\mathbf{X}_k = (\mathbf{X}_{\boldsymbol{L}_k}, \mathbf{X}_{\mathbf{G}_k})$$
$$\mathbf{X}_{\boldsymbol{L}_k} = (\mathbf{X}_{\boldsymbol{I}_k}, \mathbf{B}_k, \mathbf{X}_{\boldsymbol{C}_k})$$
$$\mathbf{X}_{\boldsymbol{I}_k} = \begin{bmatrix} {}^L\mathbf{R}_{I_k} & {}^L\mathbf{p}_{I_k} & {}^L\mathbf{v}_{I_k} \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix}, \mathbf{B}_k = \begin{bmatrix} \mathbf{b}_{\omega_k}^\top & \mathbf{b}_{a_k}^\top \end{bmatrix}^\top$$
$$\mathbf{X}_{\boldsymbol{C}_k} = (\mathbf{X}_{CP_{k_k}}, \cdots, \mathbf{X}_{CP_{k-c+1_k}}), \mathbf{X}_{CP_{j_k}} = \begin{bmatrix} {}^L\mathbf{R}_{I_{j_k}} & {}^L\mathbf{p}_{I_{j_k}} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix}$$
$$\mathbf{X}_{\boldsymbol{G}_k} = \begin{bmatrix} {}^L\mathbf{R}_{G_k} & {}^L\mathbf{p}_{G_k} & {}^L\mathbf{p}_{f_{1_k}} & \cdots & {}^L\mathbf{p}_{f_{K_k}} \\ \mathbf{0}_{1\times3} & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{1\times3} & 0 & 0 & \cdots & 1 \end{bmatrix} \tag{106}$$

The error state keeps the same formulation as (5). When the feature is associated with the pose clone, the state definition becomes:

$$\mathbf{X}_k = (\mathbf{X}_{\boldsymbol{I}_k}, \mathbf{B}_k, \mathbf{X}_{\boldsymbol{C}_k}, \mathbf{X}_{\boldsymbol{G}_k})$$
$$\mathbf{X}_{\boldsymbol{I}_k} = \left[\begin{array}{c|cc} {}^L\mathbf{R}_{I_k} & {}^L\mathbf{p}_{I_k} & {}^L\mathbf{v}_{I_k} \\ \hline \mathbf{0}_{2\times 3} & \multicolumn{2}{c}{\mathbf{I}_{2\times 2}} \end{array}\right]$$
$$\mathbf{B}_k = \begin{bmatrix} \mathbf{b}_{\omega_k}^\top & \mathbf{b}_{a_k}^\top \end{bmatrix}^\top \tag{107}$$
$$\mathbf{X}_{\boldsymbol{C}_k} = (\mathbf{X}_{CP_{k_k}}, \cdots, \mathbf{X}_{CP_{k-c+1_k}}),\ \mathbf{X}_{CP_{k_k}} = \left[\begin{array}{c|cc} {}^L\mathbf{R}_{I_{j_k}} & {}^L\mathbf{p}_{I_{j_k}} & {}^L\mathbf{p}_{f_k} \\ \hline \mathbf{0}_{2\times 3} & \multicolumn{2}{c}{\mathbf{I}_{2\times 2}} \end{array}\right]$$
$$\mathbf{X}_{\boldsymbol{G}_k} = \begin{bmatrix} {}^L\mathbf{R}_{G_k} & {}^L\mathbf{p}_{G_k} \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix} \tag{108}$$

And the feature error state is formulated as:

$${}^L\mathbf{p}_{f_k} = \mathrm{Exp}(\boldsymbol{\xi}_{\boldsymbol{R}_{CP_{k_k}}}) {}^L\widehat{\mathbf{p}}_{f_k} + \mathbf{J}_l(\boldsymbol{\xi}_{\boldsymbol{p}_{CP_{k_k}}}) \boldsymbol{\xi}_{\boldsymbol{p}_{Lf_k}} \tag{109}$$

B. State Propagation

The derivation of state propagation is same as that in Section. A regardless of whether the feature point is associated with the relative transformation or the pose clone.

C. State Update

When the feature point is associated with the relative transformation, all the state update equations are same as that in MSC-RIEKF-FDR. When the feature point is assoicated with the pose clone, all the state update equations are same as that in MSC-RIEKF-FC.

D. Observability Analysis

When the feature point is associated with the relative transformation, the observability analysis is same as that of MSC-RIEKF-FDR. When the feature point is associated with the pose clone, the observability analysis is same as that of MSC-RIEKF-FC. In this way, the dimension of the unobservable subspace is well reserved and the problem of computation cost in MSC-RIEKF-FC is solved.

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